

RESPONSIBLE ARTIFICIAL INTELLIGENCE

Unit 2: Ethics of Data Power, Consent, and Digital Inequality

CIA 1 — Component 2

AI-Integrated Responsible Data Audit using Generative AI

Platform Audited: Strava

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Responsible Artificial Intelligence

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Section 1: Introduction

- **Application Selected:** Strava — a GPS-based fitness-tracking and social-networking platform used by runners, cyclists, and other athletes.
- **Reason for Selection:** Strava sits at the intersection of several categories in the assignment brief — it is simultaneously a social platform (activity feeds, kudos, comments, leaderboards) and a data-intensive productivity/health tool. Its business model depends on continuous geolocation and biometric data collection, and it has been at the centre of well-documented, real-world privacy incidents, making it a rich case for studying data power, consent, and digital inequality.
- **Platform Description:** Strava lets users record runs, rides, swims, and other activities using GPS, connect wearables (Garmin, Apple Health, Peloton, heart-rate monitors), and share this activity with a public or semi-public social network. It offers a free tier and a paid 'Strava Premium/Summit' subscription that unlocks analytics, route-building, and leaderboard features.

Section 2: Privacy and Consent Analysis

2.1 Data Collected

- **Activity Data:** GPS routes, pace, distance, elevation, and timestamps for every recorded activity.
- **Health & Biometric Data:** heart rate, heart-rate variability (HRV), VO2max, sleep, and step count, pulled from connected devices and apps such as Garmin, Apple Health, or Peloton.
- **Profile & Account Information:** name, email, profile photo, bio, and preferences, including data received when signing in via Google or Apple.
- **Social Content:** comments, kudos, photos, club memberships, and posts.
- **Usage and Device Data:** app interaction data, device identifiers, and location metadata beyond activity recording.

2.2 Purpose of Collection

Strava states it uses this data to operate and improve the Services, power features such as route recommendations and AI-based leaderboard-cheating detection, personalize the social experience, and — for aggregated/de-identified activity data — to sell access through 'Strava Metro' to city planners and government agencies for transportation planning.

2.3 Consent Mechanisms

Consent is largely bundled into acceptance of the Terms of Service at sign-up, which is a single, one-time, all-or-nothing action. Strava does layer some granular, opt-in consent on top of this baseline: health data from connected devices is only collected when a user actively links that device, and it is not sold or used for advertising. However, most other categories (location visibility, activity sharing, third-party app data) are governed by default privacy settings that a user must proactively discover and adjust after signup, rather than by an upfront, itemized choice.

2.4 User Rights and Controls

- **Access & Portability:** users can view and export their data (including activity files) through account settings and a Bulk Data Export Tool.

- **Correction:** profile and account information can be edited directly by the user.
- **Deletion & Withdrawal:** users may request removal of previously shared personal data, and account deletion is available, though some public contributions (e.g., segments/routes) may remain visible even after account closure.
- **Additional State-Law Rights:** residents of certain U.S. states (e.g., California, Colorado, Virginia) have added rights such as opting out of the sale/sharing of personal information.

Section 3: Ethical Data Audit

3.1 Privacy Concerns

Strava collects highly sensitive geolocation and biometric data. Independent privacy reviews have scored Strava's overall privacy practices poorly, citing data retention beyond what seems necessary and sharing with advertising and analytics partners. Location data is especially sensitive: researchers have shown that only a handful of location points can re-identify a supposedly 'anonymous' individual, which undermines claims that aggregated Strava data (e.g., the Global Heatmap or Strava Metro feeds) is truly anonymous.

3.2 Manipulative Practices or Dark Patterns

A widely cited UX case study documents a Strava trial-subscription screen that used an 'Upgrade' framing alongside a soft 'decline' option, a pattern associated with confirmshaming and unclear opt-in/opt-out language. Separately, several core features (route planning, full leaderboards, detailed performance analytics, and the annual 'Year in Sport' summary) were previously free and later moved behind a paywall, a practice users have criticized as feature paywalling rather than classic informed pricing.

3.3 Data Power and Surveillance Concerns

The 2018 Global Heatmap incident, in which aggregated public activity data revealed the outlines of secret military bases, and a 2025 case in Sweden where uploaded workouts exposed the residences and travel patterns of a Prime Minister's bodyguards, both demonstrate how a platform built around 'harmless' fitness data can function as a large-scale surveillance surface once activity data is aggregated and cross-referenced. This illustrates a broader data-power imbalance: Strava (and downstream data buyers via Strava Metro) can derive population-level insight that individual users never anticipated when they agreed to share a single run.

3.4 Digital Inequality Issues

Meaningful participation in Strava's core social and competitive features (segments, leaderboards, route planning, detailed analytics) increasingly requires a paid subscription, plus a GPS watch or smartphone and reliable connectivity — resources not equally available across income levels or regions. A free-tier user experiences a materially different, more limited platform than a paying user, which runs counter to equitable access to the health and community benefits the app is marketed around.

3.5 Representational Harm Concerns

Leaderboards and public segment competitions implicitly reward users who have more leisure time, live in areas that are safe to exercise outdoors, and can afford premium equipment — factors correlated with income, gender, disability status, and geography. Users who cannot safely run outdoors, who exercise indoors, or who use lower-cost devices with less accurate GPS may be systematically under-represented or disadvantaged in Strava's social ranking systems, even though the platform presents these rankings as neutral, purely performance-based measures.

Section 4: Critical Evaluation of Generative AI

The five prompts specified in the task brief were run against a generative AI assistant. The screenshots below document each prompt and its full response. The table that follows cross-checks specific AI-generated observations against Strava's actual Privacy Policy and independent credible sources.

4.0 AI Prompt Screenshots

Prompt 1: "Summarize the privacy policy of Strava and identify the types of personal data collected."

Summarize the privacy policy of Strava and identify the types of personal data collected

Here's a plain-language summary of **Strava's Privacy Policy** (effective January 1, 2026):

Summary

Strava collects personal information to:

- Create and manage your account.
- Record and analyze your fitness activities.
- Provide social features (followers, clubs, comments, kudos, leaderboards).
- Improve existing features and develop new ones (including AI-powered features if applicable).
- Process payments for subscriptions.
- Keep the platform secure, prevent fraud, and comply with legal obligations.
- Personalize your experience and, where permitted, provide advertising and marketing. [Strava v1](#)

The company also provides privacy controls that allow users to:

- Control who can view their profile and activities.
- Create privacy zones to hide sensitive locations (such as home or work).
- Opt out of contributing data to features like the Global Heatmap, Strava Metro, and some product improvement initiatives.
- Download or delete their personal data. [Strava Help Center v1](#)

Types of personal data collected

Strava may collect the following categories of personal data:

Category	Examples
Account information	Name, email address, username, password, athlete ID, date of birth
Profile information	Profile photo, gender (if provided), height, weight, biography
Fitness and health data	Activities, workout history, heart rate, cadence, power, calories burned, exercise metrics (with consent where required)
Location data	GPS routes, start/end locations, speed, direction, elevation, map data
Device and technical information	IP address, browser type, operating system, device identifiers, app usage logs
Social data	Followers, following lists, clubs, comments, likes ("kudos"), uploaded photos and videos
Payment information	Subscription purchases and billing information (payment processing is handled by payment providers)
Communications	Messages with customer support, survey responses, feedback
Usage and analytics	Features used, interactions with the app, cookies and similar tracking technologies
Inferences	Fitness trends, preferences, estimated performance metrics, and other insights generated from your activity data

Data sharing

Strava states that it may share information with:

- Service providers (hosting, analytics, payment processors, customer support).
- Connected third-party applications that you authorize.
- Business partners and advertising partners where applicable.
- Law enforcement or government authorities when legally required.
- Third parties using aggregated or de-identified data for community features or analytics. [Strava v1](#)

Important privacy notes

- Health-related data (such as heart rate) is collected only with your consent where required.
- Public activities and routes may be visible to others depending on your privacy settings.
- You can change activity visibility, configure privacy zones, and opt out of certain data uses through Strava's privacy controls.
- Public segments, routes, and some community contributions may remain available even after an account is deleted. [Strava Help Center v2](#)

[Sources](#)

Prompt 2: "Analyze whether the consent mechanisms in Strava follow ethical principles of informed consent."

Analyze whether the consent mechanisms in Strava follow ethical principles of informed consent

Strava's consent mechanisms generally align with many principles of informed consent, but there are also ethical concerns. Here's an analysis based on common ethical principles.

Ethical Principle	How Strava Performs	Assessment
Disclosure	Strava explains what data it collects, why it is collected, how it is used, and who it is shared with in its Privacy Policy.	Mostly Meets – Information is available but spread across lengthy legal documents, making it difficult for many users to fully understand.
Comprehension	The policy is written in legal and technical language. Although Strava provides some help articles and summaries, many users may not read or understand all details.	Partially Meets – Information is present but not always easy for the average user to comprehend.
Voluntary Consent	Users choose whether to create an account and can connect or disconnect third-party apps. Some data processing is optional, while essential processing is required to use the service.	Mostly Meets – Consent is generally voluntary, though users must accept core data collection to access the platform.
Specific Consent	Permissions such as GPS location, heart-rate data, camera access, and third-party integrations are requested separately through the app or operating system where applicable.	Meets – Sensitive permissions are usually requested individually rather than bundled together.
Right to Withdraw	Users can change privacy settings, revoke permissions, disconnect third-party services, download their data, or delete their account.	Meets – Users have multiple ways to withdraw consent or limit data sharing.
Transparency	Strava explains data sharing with service providers, connected apps, and legal authorities. It also describes privacy controls such as Privacy Zones and activity visibility.	Mostly Meets – Transparency is good, though some data uses (analytics, personalization, or AI-related features) may not be obvious to all users.

Ethical Strengths

- Provides user controls for activity visibility and location privacy.
- Allows users to download or delete their personal data.
- Requests separate permissions for sensitive device features such as GPS and health sensors.
- Clearly identifies major categories of data collected and purposes for collection.

Ethical Weaknesses

- **Lengthy Privacy Policy:** Many users are unlikely to read or fully understand the policy before consenting.
- **Complex Language:** Legal terminology can make informed decision-making difficult.
- **Bundled Acceptance:** Users cannot use most core services without agreeing to essential data collection, limiting practical choice.
- **Potential Information Overload:** Important privacy details may be buried in long documents instead of being highlighted at the moment users make decisions.

Overall Evaluation

From an ethical perspective, Strava largely follows the principles of informed consent, particularly by providing privacy controls, allowing users to withdraw consent, and requesting permissions for sensitive data. However, it does not fully satisfy the ideal of informed consent because the privacy policy is long and complex, making it difficult for many users to understand exactly how their personal data will be collected and used.

Conclusion: Strava demonstrates moderate to strong ethical compliance with informed consent principles. It performs well in transparency and user control but could improve by presenting privacy information in simpler language, using concise summaries, and offering clearer, just-in-time explanations when users are asked to share sensitive data.

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Prompt 3: "Identify any manipulative data practices, dark patterns, or privacy concerns in Strava."

Identify any manipulative data practices, dark patterns, or privacy concerns in Strava.		
Strava includes a number of privacy features, but like many social fitness platforms, it also raises ethical concerns related to data collection, user behavior, and interface design. Here's an objective analysis.		
Area	Potential Concern	Impact
Default sharing settings	New users may unintentionally share activities publicly or with a wider audience than expected if they do not review privacy settings.	May expose home, workplace, or daily routines.
Location tracking	Strava continuously collects precise GPS data for activities. Even with Privacy Zones, repeated routes may reveal sensitive locations or personal habits.	Risk of stalking, profiling, or physical safety concerns.
Social pressure and gamification	Features such as leaderboards, Kudos, achievements, and challenges encourage users to share more activities and spend more time on the platform.	Can influence users to disclose more personal information than they otherwise would.
Complex privacy settings	Privacy controls exist but are spread across multiple menus and options. Some users may not fully understand how visible their activities are.	Users may believe their data is private when it is not.
Third-party integrations	Connecting fitness devices and apps allows data to flow between multiple services. Users may not fully understand how external apps use their information.	Greater risk of data sharing beyond Strava.
Extensive data collection	Strava collects account details, location, activity metrics, device information, and interaction data to improve services and personalize experiences.	Creates detailed behavioral and fitness profiles.
Possible Dark Patterns		
While Strava is generally considered more transparent than many social media platforms, some interface choices could be viewed as mild dark patterns:		
Privacy settings not always emphasized - Users often focus on recording activities rather than reviewing privacy options. - Important controls may require navigating several settings pages. Gamification encouraging disclosure - Badges, challenges, leaderboards, and social recognition encourage users to post more workouts and route information. - This benefits community engagement but may reduce thoughtful privacy decisions. Encouraging account linking - Users are prompted to connect wearables and third-party fitness services. - The benefits are clear, but the implications for data sharing may receive less attention. Long privacy documents - Consent is obtained through lengthy policies that most users are unlikely to read completely. - This weakens the practical effectiveness of informed consent.		
Privacy Concerns		
Sensitive location information: GPS routes can reveal home addresses, workplaces, schools, or daily routines if privacy settings are not configured carefully. Health and fitness data: Heart rate, exercise habits, and performance metrics are sensitive personal information that could be valuable to advertisers or other third parties if misused. Behavioral profiling: Combining activity history, social interactions, and device data can create detailed profiles of users' lifestyles and habits. Third-party data sharing: Users who authorize integrations may expose their information to external services with different privacy practices. Historical risks: In 2018, Strava's public Global Heatmap inadvertently revealed the locations of some military bases because users' public activity data was aggregated. This highlighted how even anonymized or aggregated location data can create unexpected privacy risks.		
Ethical Assessment		
Positive practices		
- Offers Privacy Zones to hide sensitive locations. - Lets users control activity visibility. - Allows users to download and delete their data. - Provides options to disconnect third-party applications.		
Areas for improvement		
- Make private activity settings the default for new users. - Present simpler, shorter explanations of data collection at the point of consent. - Highlight the privacy implications of GPS sharing before users upload activities. - Make third-party data sharing more explicit and easier to review.		
Conclusion		
Strava does not appear to rely on highly deceptive dark patterns, but it does employ engagement-focused design choices that can encourage users to share more data than they may realize. The primary privacy concerns stem from precise location tracking, extensive collection of fitness and behavioral data, and the complexity of privacy settings rather than overtly manipulative consent mechanisms.		

Prompt 4: "Evaluate potential issues of digital inequality and representational harm associated with data collection practices in Strava."

Evaluate potential issues of digital inequality and representational harm associated with data collection practices in Strava.

Strava's data collection practices provide valuable fitness insights, but they can also contribute to digital inequality and representational harm. These issues relate to who benefits from the platform, whose data is represented, and who may be disadvantaged.

1. Digital inequality

Digital inequality refers to unequal access to technology, digital services, or the benefits derived from data.

Issue	Explanation	Example
Access to technology	Strava works best with smartphones, GPS devices, and smartwatches, which may be expensive.	People from lower-income backgrounds may not have access to premium fitness devices or subscriptions.
Internet dependence	Users need reliable internet to sync activities and use social features.	Rural users or those with poor connectivity may have limited access.
Premium subscription	Advanced analytics and training features require a paid subscription.	Wealthier users receive more insights and personalized recommendations than free users.
Digital literacy	Managing privacy settings requires users to understand data-sharing options.	Less tech-savvy users may unintentionally expose personal information.
Global representation	Most Strava users come from regions with high smartphone adoption.	Developing countries and underserved communities are underrepresented in the platform's data.

Impact of Digital Inequality

- Fitness insights and recommendations may primarily reflect users from wealthier populations.
- Public datasets such as activity heatmaps may overlook areas where fewer people use Strava.
- Researchers or city planners using Strava-derived data could make decisions that favor already well-represented communities.

2. Representational Harm

Representational harm occurs when collected data does not accurately or fairly represent all groups or reinforces existing biases.

Issue	Explanation	Example
Sampling bias	Strava users are not representative of the general population.	Older adults, children, and low-income populations are less likely to appear in the data.
Urban bias	More activities are recorded in cities than rural regions.	Heatmaps may suggest urban areas need more cycling infrastructure while overlooking rural needs.
Fitness bias	Active athletes generate more data than casual exercisers.	Health research based solely on Strava may overestimate average activity levels.
Gender and demographic imbalance	Some demographic groups may be less represented or have different sharing behaviors.	Training models built on Strava data could produce recommendations that work better for dominant user groups.
Disability representation	People with disabilities may not be adequately represented in activity datasets.	Accessibility needs could be overlooked in analyses based on Strava data.

Impact of Representational Harm

- AI models trained on Strava data may produce recommendations that are less accurate for underrepresented groups.
- Public health studies using Strava data may not reflect the activity patterns of the broader population.
- Infrastructure planning based on Strava activity could unintentionally reinforce inequalities by directing resources toward already active or affluent areas.

Ethical Implications

Fairness

The benefits of Strava's analytics and community features are not equally distributed. Users with better devices, stronger internet access, and paid subscriptions gain more value.

Inclusivity

The platform's datasets are skewed toward technologically connected and fitness-oriented individuals, making them less representative of society.

Justice

If governments or researchers rely heavily on Strava data for planning or policy decisions, communities that are underrepresented in the data may receive less attention or investment.

Recommendations

To reduce digital inequality and representational harm, Strava could:

- Make advanced privacy settings simpler and more accessible.
- Offer more useful analytics in the free version to reduce economic barriers.
- Clearly communicate that Strava data is not representative of the entire population.
- Encourage participation from diverse demographic groups.
- Evaluate algorithms for demographic and geographic bias before deploying new features.
- Collaborate with public health organizations to account for underrepresented populations when using aggregated activity data.

Conclusion

Strava's data collection is highly valuable for fitness tracking and research, but it also reflects the characteristics of its user base rather than the population as a whole. This creates digital inequality by favoring users with greater technological and financial resources, and representational harm by underrepresenting certain demographic and geographic groups. Ethical use of Strava data therefore requires careful consideration of these biases, especially when the data informs research, AI systems, or public policy decisions.



Prompt 5: "List ethical concerns that may arise from the platform's data practices from a Responsible AI perspective."

List ethical concerns that may arise from the platform's data practices from a Responsible AI perspective.

From a **Responsible AI** perspective, Strava's data practices raise several ethical concerns related to fairness, privacy, transparency, accountability, and user autonomy.

Responsible AI Principle	Ethical Concern	Explanation
Privacy	Collection of sensitive personal data	Strava collects GPS locations, fitness activities, heart rate, device information, and social interactions. If not properly protected, this data could expose users' daily routines or health information.
Transparency	Complex privacy policies	Users may not fully understand what data is collected, how it is processed, or who it is shared with because privacy notices can be lengthy and technical.
Informed Consent	Consent may not be fully informed	Although users agree to the privacy policy, many may not read or fully understand it, making consent less meaningful.
Fairness	Biased datasets	AI systems trained on Strava data may favor groups that are overrepresented (e.g., urban, affluent, or highly active users), leading to less accurate recommendations for others.
Inclusiveness	Underrepresentation of certain groups	Older adults, people with disabilities, rural communities, and lower-income populations may contribute less data, reducing the inclusiveness of AI models.
User Autonomy	Gamification influencing behavior	Leaderboards, badges, challenges, and social recognition may encourage users to share more personal information or exercise in ways driven by rewards rather than personal choice.
Accountability	Limited visibility into AI decisions	Users may not know how recommendation algorithms, route suggestions, or performance insights are generated or how their data influences these outputs.
Security	Risk of data breaches	Large volumes of sensitive location and health data require strong security measures. A breach could expose highly personal information.
Purpose Limitation	Secondary use of data	Data collected for fitness tracking could later be used for analytics, product improvement, or other purposes that users did not originally anticipate.
Data Minimization	Extensive data collection	Collecting more information than necessary increases privacy risks and the potential impact of misuse or unauthorized access.

Additional Responsible AI Concerns

- Location Privacy**
 - Continuous GPS tracking can reveal home addresses, workplaces, travel habits, and daily routines.
 - Even anonymized location data may sometimes be re-identified.
- Algorithmic Bias**
 - AI models trained on non-representative datasets may generate recommendations that perform better for some users than others.
- Profiling**
 - Combining fitness, behavioral, and social data can create detailed personal profiles that may influence personalization or advertising.
- Third-Party Data Sharing**
 - Users may authorize integrations without fully understanding how external services collect, retain, or process their data.
- Risk of Misuse**
 - Aggregated fitness data could be used in ways users did not expect, such as research, infrastructure planning, or commercial analytics, potentially affecting communities differently.

Recommendations for Responsible AI

To better align with Responsible AI principles, Strava could:

- Use **plain-language privacy notices** and concise summaries.
- Make **privacy-protective settings the default** for new users.
- Provide **clear explanations of AI-generated recommendations** and how user data contributes to them.
- Conduct **regular bias and fairness audits** on recommendation algorithms.
- Apply **data minimization**, collecting only the information necessary for each feature.
- Strengthen **user control**, allowing people to easily opt in or out of specific types of data collection and sharing.
- Improve **security safeguards** and communicate them clearly to users.

Conclusion

From a Responsible AI perspective, the main ethical concerns in Strava's data practices are:

- Privacy and protection of sensitive health and location data.
- Meaningful informed consent.
- Transparency about data use and AI decision-making.
- Fairness and representativeness of AI models.
- User autonomy in the face of engagement-driven design.
- Accountability for AI outcomes.
- Strong security and responsible data governance.

Addressing these issues would help ensure that Strava's AI-driven features remain trustworthy, fair, and respectful of users' rights while reducing the risk of harm.

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4.1 Verification Table

AI Observation	Correct/ Incorrect	Evidence from Privacy Policy	Missing Ethical Issues	AI Limitations/Halluc inations
Prompt 1 response stated users can opt out of contributing data to the Global Heatmap and Strava Metro, and can create 'Privacy Zones' to hide sensitive locations like home or work.	Correct	This matches Strava's publicly documented privacy controls (added after the 2018 heatmap incident) and its Data & Privacy support page, which describes location-hiding and data-use opt-outs.	The AI did not mention that Privacy Zones only obscure the start/end points of a route — the rest of the route, and any earlier activities recorded before the zone was set up, may still be exposed.	None found, but the response slightly overstates how complete this protection is.
Prompt 2 response rated 'Specific Consent' as fully met, stating sensitive permissions (GPS, heart rate, camera) are 'usually requested individually rather than bundled together.'	Partially Correct	Device-level permissions (GPS/camera) are indeed separate OS prompts. However, Strava's own Privacy Policy shows broader data-use purposes — service improvement, personalization, AI features, marketing — are all accepted in one bundled Terms-of-Service click at sign-up, not granular purpose-by-purpose.	The AI did not distinguish device-permission consent (granular) from platform-purpose consent (bundled), which is the more significant informed-consent gap the course unit focuses on.	The 'Meets' rating for Specific Consent is generous — it reflects OS-level design, not Strava's own consent architecture.
Prompt 3 response identified 'default sharing settings' as a mild dark pattern, warning new users may unintentionally share activities more widely than expected.	Correct	This concern is corroborated by independent privacy reviews rating Strava's overall data practices poorly and by recurring user reports of unexpectedly public activity data.	The AI did not name this using the standard dark-pattern taxonomy (e.g., 'default framing' bias), nor connect it to the fact that Strava's Terms explicitly warn users in sensitive roles to manage geolocation sharing carefully — implying Strava itself knows the default settings carry risk.	None found — the claim is accurate but under-specified.

Prompt 3 response cited the 2018 Global Heatmap incident, where aggregated public activity data revealed the outlines of military bases.	Correct	Confirmed by independent legal/academic sources (Journal of High Technology Law), which also document a 2025 Sweden incident exposing security personnel's residences and travel routes via Strava uploads.	The AI did not connect this to digital inequality — the users able to get dedicated protection guidance afterward were institutional/high-profile (military, VIP security), while ordinary users received the same generic privacy settings.	None found on the facts, but the equity angle is missing.
Prompt 4 response stated that advanced analytics and training features require a paid subscription, so 'wealthier users receive more insights and personalized recommendations than free users.'	Correct	Confirmed by industry research (Contrary Research), which documents route planning, full leaderboards, and detailed performance analysis being moved behind Strava's premium paywall, with only discounted student/family plans as concessions.	The response did not mention that some of these features (e.g., 'Year in Sport,' route planning) were previously free and later paywalled — a distinct practice from designing a paid tier from the outset, and arguably a stronger inequality concern since it took away access users already had.	None found — accurate but could be more specific about the paywalling timeline.
Prompt 4 response identified 'sampling bias' — Strava users are not representative of the general population, so heatmaps and public datasets may overlook areas where fewer people use the app.	Correct	This matches well-documented critiques of Strava data used in cycling/urban-planning contexts, where infrastructure decisions based on Strava Metro data risk favoring neighborhoods that are already more affluent and active.	The AI's own recommendation ('communicate that Strava data is not representative') is reasonable, but it did not flag that Strava sells this same data through Strava Metro to governments for infrastructure planning — the exact context where sampling bias causes real-world representational harm.	None found — the concern is accurate but Strava Metro's role was left out.

Prompt 5 response listed 'Purpose Limitation' as a concern, noting fitness-tracking data could later be used for analytics, product improvement, or other purposes users did not originally anticipate.	Partially Correct	Strava's Privacy Policy does state data is used to 'provide, improve, and develop features,' which supports this claim, but the AI did not name the most concrete example: sale of aggregated/de-identified activity data to third parties via Strava Metro, which is a documented, specific secondary use.	Missing the direct link to Strava Metro as the clearest real-world example of purpose expansion beyond a user's original fitness-tracking intent.	The concern is stated abstractly/generically rather than tied to Strava's actual, named data product — a common AI limitation of staying at a general level unless prompted for specifics.
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4.2 Correct Observations Generated by AI

Across all five prompts, the AI accurately identified the main categories of data collected (location, activity, health/biometric, profile, and social data), correctly described Strava's Privacy Zone and heatmap/Metro opt-out controls, correctly rated most consent mechanisms as reasonable but imperfect, and correctly surfaced the real, independently documented 2018 Global Heatmap incident as evidence of surveillance risk. The digital-inequality and representational-harm analysis (Prompt 4) was particularly strong, correctly naming sampling bias, urban bias, and paywall-driven inequality without being explicitly asked to.

4.3 Missing Ethical Concerns

Across multiple prompts, the AI under-addressed Strava Metro — the commercial channel through which aggregated/de-identified activity data is sold to governments and urban planners. This is the single clearest example connecting purpose limitation (Prompt 5), sampling bias (Prompt 4), and data-power concerns (Prompt 3) in Strava's actual business model, yet the AI treated each of these as separate, generic concerns rather than tracing them back to one concrete data product.

4.4 Hallucinations or Incorrect Claims

No outright fabrications were found, but Prompt 2's response rated 'Specific Consent' as fully 'Meets' based on OS-level device permissions (GPS, camera) rather than Strava's own platform-level consent design, which is actually bundled into a single Terms-of-Service acceptance. This is a subtle but meaningful overstatement of how informed Strava's consent process really is.

4.5 Biases and Limitations of the AI Tool

- Recency limitations: the AI tool's training data may predate the January 2026 rewrite of Strava's Terms and Privacy Policy, so live verification against the current policy text was necessary.
- Surface-level ethical framing: without prompting for specific frameworks (e.g., data-power theory, algorithmic fairness), the AI defaulted to generic 'privacy risk' language rather than analyzing structural inequality in depth.
- Tendency to assign intent: the AI sometimes framed ambiguous design choices as deliberately manipulative rather than presenting alternative explanations (e.g., poor UX vs. dark pattern).

- No independent access to the live app: the AI cannot browse the current in-app UI itself, so claims about interface design rely entirely on secondary sources rather than direct observation.

Section 5: Ethical Reflection (250–300 words)

Auditing Strava revealed how easily 'fitness' data can double as surveillance data. A single GPS track feels harmless in isolation, but aggregated across millions of users it can expose military installations, the homes of public figures, and the daily routines of ordinary people — showing that data power lies less in any one data point than in the ability to combine and analyze data at scale, a capability only the platform itself possesses.

This raises real doubts about whether users can give truly informed consent online. Strava's consent model asks for one broad agreement at sign-up, long before a user can anticipate how their data might later be aggregated, sold through Strava Metro, or exposed through a public heatmap. Consent given without understanding downstream aggregation risk is consent in name only, and Strava's more genuine, granular consent step is reserved for health-device syncing, not for the location and social-visibility settings where the biggest risks actually lie.

Organizations, in my view, carry a responsibility that goes beyond legal compliance: they should design defaults that protect the most vulnerable users automatically, rather than requiring everyone to locate and adjust privacy settings themselves, and they should be transparent about secondary revenue uses of 'anonymized' data given how easily location data can be re-identified.

Finally, this exercise showed both the value and the limits of generative AI for ethical auditing. The AI tool was fast and comprehensive at summarizing stated policy, but it under-explored digital inequality and occasionally implied intent that the source evidence did not fully support. Generative AI is therefore best used as a first-pass research assistant for this kind of audit, not as a substitute for verifying claims against the actual privacy policy and independent, credible reporting.

Section 6: References

- Strava, Inc. Privacy Policy (effective January 1, 2026). <https://www.strava.com/legal/privacy>
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